Administrating MANETs Via Brain Computer Interface

N. Chandrakant

Abstract The electroencephalogram (EEG) based Brain-Computer Interfaces (BCIs) has been considerably extended to the multidisciplinary and challenging nature of BCI study. The algorithms of Digital signal processing and pattern recognition certainly constitute necessary components of a BCI system which are applied to the EEG signals to decode mental states those are relevant for BCI operation. The basic BCI concepts such as: brain activity monitoring, BCI operation, and the relevant mental states for BCI are applied to WSN. The major types of relevant mental states for BCI, namely motor imagery (ERD/ERS), steady state visual evoked potentials (SSVEP) are also mapped to MANET's behaviour. The multivariate nature of the EEG combined with the neuroscience familiarity on hemispheric brain specialization is favourably taken into account to derive optimal combinations of the individual signals composing the EEG. BCI provides a non-muscular channel for sending commands to the external world using the electro-encephalographic activity or other electrophysiological process of the brain function. A necessary factor in the successful operation of BCI systems is the methods used to process the brain signals. The processed data has been applied on to nodes of MANET where each set of sensor nodes are used for particular application. Based on the output of BCI commands, the MANET is operated and controlled with using only activities of neurons of brain, hence this approach could be more helpful to physically differently abled people or medically injured people.

Keywords WSN · BSP · ANN · BCI (Brain Computer Interface)

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1 Introduction

A Brain Computer Interface (BCI)/Mind Machine Interface (MMI)/Direct Neural Interface/Brain Machine Interface (BMI)/Direct Communication Pathway between the brain and an external digital device which are often directed at assisting, augmenting, or repairing human cognitive or networking functions, sensory-motor functions [1].

Human brains are made up of neurons with entity nerve cells connected to one another by dendrites and axons. Whenever we think, move, feel or remember something, our neurons are at work which is carried out by small electric signals that zip from neuron to neuron as fast as 250 mph. The signals are produced by differences in electric potential carried by ions on the covering of each neuron. We can analyse the signals and process as understandable words.

The applications of BCI can include,

(a) You can sense the thoughts of Aliens if scientists proves its existence.
(b) Benefits for stroke.
(c) Benefits for disabilities like paralyzed including oral problems.
(d) Study of Animal's behaviour.
(e) Identify thoughts of thieves.
(f) BCI system will facilitate an augmentative communication method for patients with severe motor disabilities.
(g) Monitor unmanned areas.

The rest of this paper is arranged as follows: In Sect. 2, gives some helpful input information for this paper. Detailed design and its implementation with results has been explained in Sect. 3. Finally, Sect. 4 concludes the paper and gives future trends.

2 Literature Survey

Starlab Neuroscience [1, 2] Research offers advanced data analysis services for their clients in research and in industry. They have a sister company Neuroelectrics, where hey offer custom solutions for data capture and analysis based on clients needs.

Intendix [3] provides 3 major solution areas, they are, Personal EEG Based Spelling System, Screen Overlay Control Interface and Paint by Thoughts. This system can work with passive, active, or new dry EEG electrodes. Using these systems, apart from writing text, the patient can also use the system to trigger an alarm, let the computer speak the written text, print out or copy the text into an e-mail or to send commands to external devices.

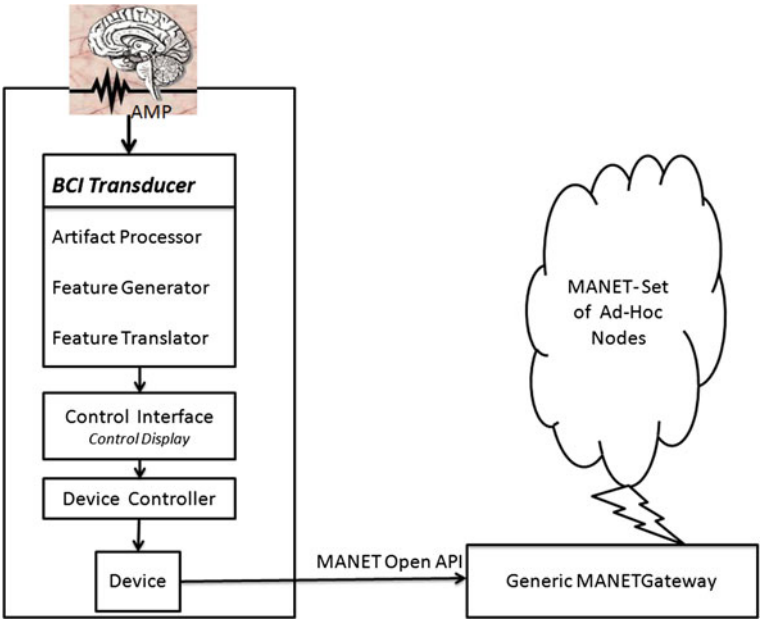


Fig. 1 Overall architecture of BCI-MANET computing

Algorithm 1 $main(args[], addlInfo[])$

Require: Initialize $i \leftarrow 0, j \leftarrow 0, rule[] \leftarrow null, n \leftarrow numberOfNodes$

```
while i++ <= numberOfRules do
  if args[i] ==' BCI_Value' then
    rule = getInstructions(addlInfo[i])
    while j++ <= n do
      if rule[i].nodeId == j then
        if rule[i] == isAlreadyApplied() then
          Then ignore this instructions and donot execute on MANET.
        else
          executeOperationOnNode(rule[i], j)
        end if
      end if
    end while
  else
    Take the default behaviour of the network/algorithm
  end if
end while
```

Paper [4] proposed a new multi-stage procedure for a real time BCI which allows a BCI [5] user to movement including four directions and to stop a small car (any object) on the computer display screen in real time. The results shows high performance of the proposed on-line BCI system which has conducted extensive experiments with 5 young healthy subjects.

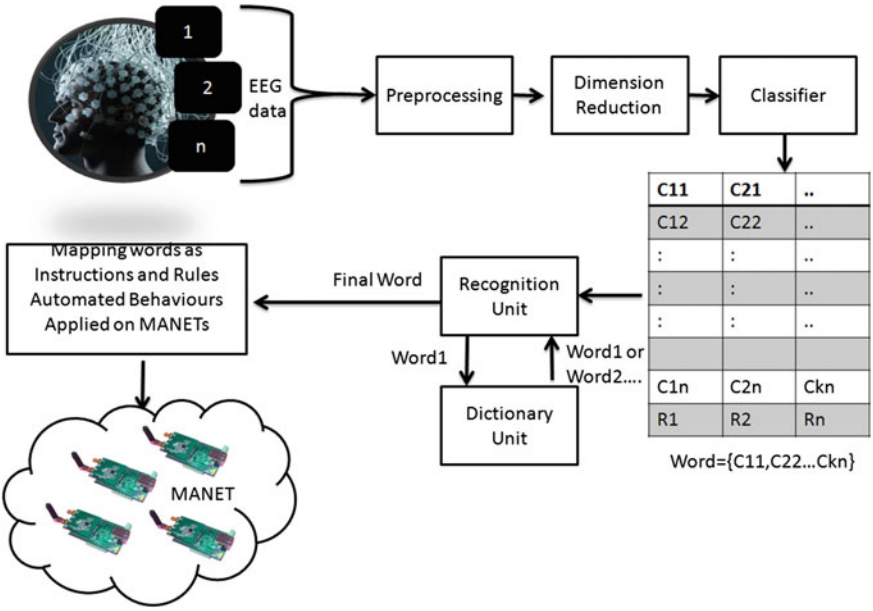


Fig. 2 Architecture of proposed system

Paper [6] presents a new classification technique of uninterrupted EEG recordings based on a network of spiking neurons. The EEG signals were pre-processed using Wavelet Transform to take out the noise and to extract the low frequency substance and analysis of the signals was carried out on the ensemble EEG and the task of the neural network was to recognize the P300 component in the signal. The network engaged leaky-integrate-and-fire (LIF) neurons as nodes in a multi-layered arrangement which does formation of multiple weak classifiers to perform voting and combined results are used for final classification.

3 Proposed System and Design

The overall Architecture is shown in Fig. 1. A transducer device will converts variations in a physical quantity, such as pressure or brightness into an electrical signal or vice versa.

Electrodes of Electro Encephalo Graph (EEG) will read the brain signals after placing electrodes in the specific area of the brain where the suitable signals are generated. The electrodes compute minute differences in the voltage between neurons and this signal is further amplified, filtered and interpreted by a computer program.

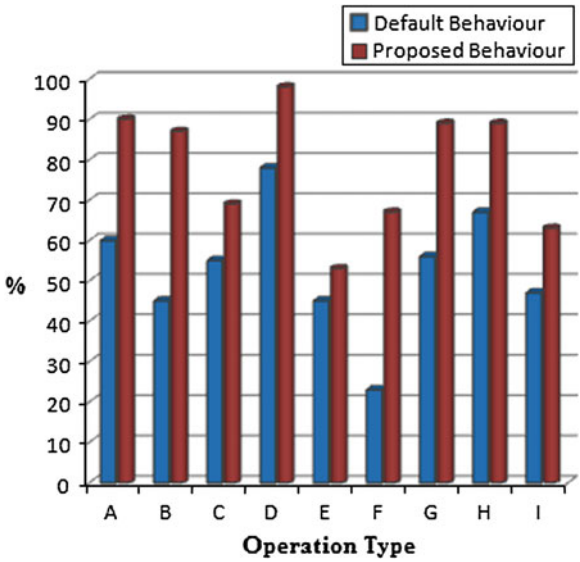


Fig. 3 Performance of default and proposed behaviour of a MANET

The proposed system is shown in Fig. 2. Electroencephalography (EEG) is the sensing of electrical action along the scalp which measures the voltage fluctuations resulting from ionic current flows between the neurons of the brain. In the medical context, EEG records the brain's spontaneous electrical activity over a short period of time, typically 20–40 min, as recorded from numerous electrodes placed on the scalp (Fig. 3).

Then preprocessing includes, Epoching which is cutting the data into chunks referenced to stimulus presentation and Averaging which is calculating the mean value for each time-point across all epochs.

Dimension reduction is generally used to decrease the necessary training time of the classifiers with a few degree of precision lost. The dimension reduction is typically performed on either feature or electrode space.

The processing and matching of words with help of dictionary out of the EEG data can be evoked by potential components, like the P300 or the N400. This will indicate collective macroscopic activities of neurons that are reflected in the EEG as speech.

The final matching word(s) is (are) mapped to certain set of instructions and rules and intern these instructions are applied as operations on MANETs which can help network to increase its throughput/performance. Sometime human intervention gives more advantages than auto-operations. MANET operations can be invoked by above strategy/algorithm and operations are as follows,

A. Routing Path Adjustment: Detecting efficient path for packet routing. It includes even deciding central node for other node's communication. Dynamic/Hybrid path can be calculated on-demand, hence it is more efficient.

B. Topology Adjustment: The structure of the network is reconstructed based on the requirements.
C. Resource Management: Limited bandwidth available between two intermediate nodes, hence managing it efficiently. Node may have limited memory and power and thus computations need to be energy-efficient.
D. Energy Management: Energy is a very important resource of MANET, a node/network is alive till energy/power is lost. Hence, it should be preserved as much as we can by analysing its usages.
E. Traffic management: The traffic can exhaust energy of the network and lost of packets too, hence transferring of packets needs to do intelligently. It includes even link management like failures and redirecting traffic.
F. Memory Management: Memory is one of the essential resource of MANET and also its memory is very small. Hence, memory must be used intelligently.
G. Sleep/Wakeup Strategy: If a node is useful or participating in the communication or transferring repeated data, in such cases, we can toggle the nodes's status to sleep or wakeup.
H. Enabling/Disabling Security: A hierarchy of security is enabled or disabled based on cost in terms of network performance and energy consumption.
I. Miscellaneous: Unspecified or customized operations can be done on MANETs, e.g., node retirement etc.

The basic algorithm is specified in Algorithm 1. The algorithm has many variables and methods, *numberOfRules* is number of rules or instructions applied on set of nodes. *BCI_Value* is the value of braincomputer interface which is generated out of EEG data. The *getInstructions(addlInfo[i])* is the process of extracting instructions out of array data. The *isAlreadyApplied()* is used to check weather instructions are already executed on the network or not. The *executeOperationOnNode(rule[i], j)* is the process of applying the operation specified in *rule[i]* on a particular node(*j*).

The performance graph of default and proposed behaviour is shown in Fig. 3 with comparison of percentage. The simulation experiment is implemented in JAVA with 500 nodes as network size. Where operation type is termed as actions/rules/instructions to be performed on node(s). Operation Types are, Routing, Topology, Resource, Energy, Traffic, Memory, Sleep/Wakeup, Security and Miscellaneous. In this experiment, we have considered Routing operation type and we simulated 9 times for the same type which is named from *A* to *I*.

It is obvious that human intelligence is more dynamic than default behaviour. Hence, our experiment proves the same and have achieved more performance than the default behaviour. Similarly, same kind of experiment can be conducted for other operation types and we believe that human intervention always fetches more performance than automated, but everything has its own merits and demerits.

4 Conclusions

A novel approach is proposed where integration of MANETs and BCI devices are linked and commands are processed using neuron's (brain) signals. The brain's data is processed and provided as input to the nodes of MANET where each set of mobile nodes are used for different applications. Based on the output of BCI system, the MANET is operated and controlled using only activities of neurons of a brain, hence this approach could be more helpful/useful to physically differently abled people, medically injured people or even lazy people too!

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